

GenAI-Supported Creative Learning in Digital Museum Education: A Case Study of Maritime Art Painting Creation

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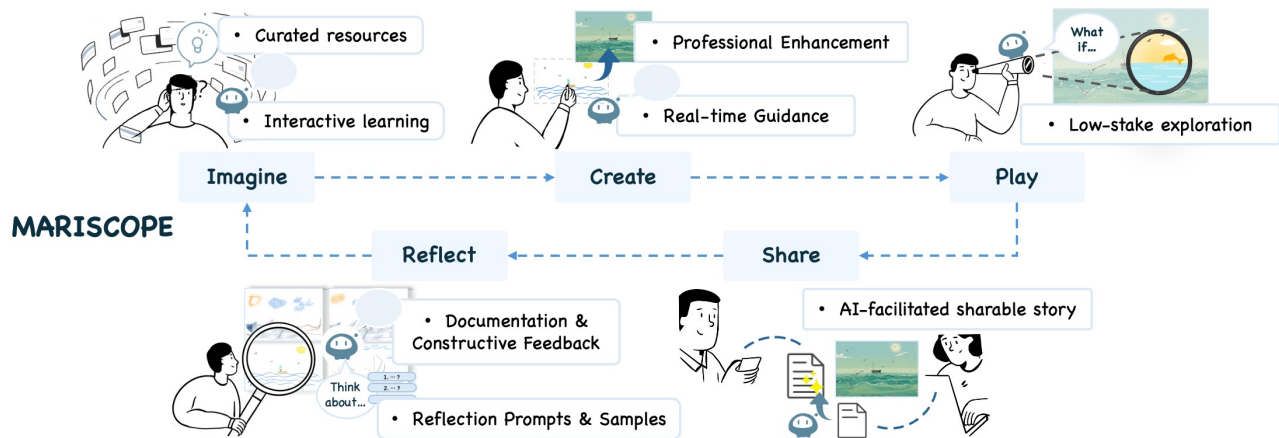


Figure 1: We leverage the generative and adaptive capacity of GenAI to support the five stages of the creative learning process, offering an enriched and iterative experience tailored to digital museum education.

Abstract

Creative learning has significantly enriched on-site museum education by fostering deeper engagement, encouraging exploration, and promoting active participation through hands-on experiences. However, the digital transformation of holistic and structured creative learning processes in museum education remains largely underexplored. Although Generative Artificial Intelligence (GenAI) holds promise for supporting creative learning, its full integration across all phases in non-formal education settings like museums remains limited. In this work, we designed Mariscope, a GenAI-supported creative learning platform for marine museum education. By integrating GenAI with the five stages of the iterative creative learning path, Mariscope offers a personalized and dynamic learning experience. A preliminary study with four participants demonstrated positive impacts on learning outcomes and creative self-efficacy, showcasing the platform's potential for enhancing digital museum

education through iterative and interactive creative learning processes.

CCS Concepts

• **Applied computing** → **Interactive learning environments**; • **Human-centered computing** → **Empirical studies in interaction design**.

Keywords

Museum education, Creative learning, AI in education, Human-AI co-creation

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1 INTRODUCTION

Museum education is a cornerstone of lifelong learning, transforming museums from passive artifact repositories into dynamic spaces for cultural, historical, and scientific engagement, fostering dialogue and creativity [19, 20, 34]. Creative learning [14], defined as an approach that emphasizes active exploration, self-expression,

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and hands-on engagement [49], significantly enhances the impact of museum education by fostering deeper engagement and promoting personalized discovery. Many museums offer on-site creative learning programs [7, 41, 42], such as workshops and art sessions, providing interactive opportunities for visitors to foster creativity and deepen engagement with themes or exhibits. However, amid the widespread proliferation of digital museums, research on the digital transformation of creative learning in museum education remains limited, even though they have the potential to enable accessible and interactive experiences. Previous research on digital creation platforms [37, 54] in museums mostly focused on content creation, with limited attention to structured, iterative, and personalized creative learning processes. Integrating complete creative learning into digital museum education requires adapting interactivity, personal guidance, and real-time feedback, aligning with the learning path [53]. Given the nonformal nature of museum education [19] and the iterative nature of the creative learning process [46], solutions should balance flexibility with structure, offering personalized guidance, real-time feedback, and inspiration in virtual environments. The rapid development of Generative AI (GenAI) technologies has revolutionized the landscape of creative tasks, achieving unprecedented growth and adoption across diverse creative domains [11, 17, 18]. In educational contexts, GenAI has demonstrated its potential to enhance creative learning by supporting problem-solving [48], fostering collaboration [32], and enabling new forms of exploration [51]. However, existing research predominantly focuses on specific aspects of creative learning, such as ideation [13, 28, 50] or creation [1, 33], often overlooking the broader, iterative nature of the creative learning process [49], leaving the full integration of GenAI to support all its phases largely unexplored. Furthermore, most studies concentrate on formal education settings, such as certain subjects in schools, with a pronounced gap in nonformal learning environments such as museums and other memory institutions, where learning is more self-directed and exploratory rather than structured and teacher-guided [6].

To address these gaps, we designed and implemented Mariscope, a GenAI-supported creative learning platform tailored for marine museum education. Mariscope harnesses the generative and adaptive capabilities of GenAI, seamlessly integrating them with the five stages of the iterative creativity cycle—Imagine, Create, Play, Share, and Reflect [46]—to foster a dynamic and personalized creative learning experience within the museum education context (Fig. 1). To assess the platform’s usability and effectiveness, we conducted a preliminary study involving four participants. The findings demonstrated positive impacts on participants’ learning outcomes and creative self-efficacy, highlighting the platform’s affordance in supporting users across various stages of the creative learning process through the strategic integration of GenAI.

2 RELATED WORK

This section reviews literature on museum education, creative learning, and applications of GenAI in educational contexts. We highlight key findings and gaps that shape our design approach.

2.1 Museum Education and Creative Learning

Museum education has long been recognized as a pivotal medium for promoting cultural understanding and lifelong learning in a rapidly evolving world [19, 20]. It serves as a bridge connecting individuals with history, art, and science in ways that deepen their appreciation for human achievements and cultural diversity [34]. Traditional museum education focused on guided tours and hands-on interactions [40], emphasizing structured learning of factual and historical contexts [26]. Today, museum education adopts creative learning to provide engaging, interactive experiences, fostering deeper connections with art, history, and science [10, 36]. Creative learning is an approach that emphasizes active engagement and exploration, encouraging learners to make meaningful connections with concepts and express themselves through hands-on, experiential activities [14, 49]. Creative learning programs in museums [7, 41, 42], use practical and collaborative activities to inspire curiosity, foster artistic expression, and encourage storytelling, helping visitors construct personal meaning and develop agency. In the current era, digital technologies have greatly expanded the reach and accessibility of museum education [5, 31], creating new opportunities for global engagement and remote learning [45], including virtual tours [30, 43, 47] and online exhibitions [9, 27, 31]. However, in current museum digitization research, we rarely see work on the digital transformation of creative learning in museum education. Previous research has explored hybrid and virtual approaches to digital creation in museum education. For instance, the Invisible Museum platform [54] enables user-designed 3D exhibitions, blending Human-Centered Design with deep learning for artifact narratives. Morse et al. explored public co-creation workshops to envision future digital museum interactivity [37]. Kang et al. introduced a Roblox-based Virtual Art Museum Education Program [22], engaging students in creating virtual museums and enhancing their knowledge through collaborative metaverse activities. However, with most studies focusing on content creation, the underlying methodologies and frameworks that facilitate the complete creative process remain unexplored, and so does research on platforms supporting structured, iterative, and personalized creative learning processes. Integrating creative learning, such as hands-on workshops, into these digital museum platforms presents unique challenges, as it requires interactive adaption and relies heavily on personal guidance, real-time feedback, and collaborative environments [53]. What’s more, museum education, by nature, is often informal, encouraging exploration, self-directed learning, and interaction in dynamic, flexible spaces [19]. This underscores the need for a solution to integrate creative learning into digital museum education while addressing the inherent challenges of providing guidance, inspiration, and structure in virtual learning environments.

2.2 GenAI Support for Creative Learning in Educational Settings

GenAI and its associated technologies, such as large language models (LLMs, e.g., GPT-4), deep learning-based image synthesis tools (e.g., DALL-E, Stable Diffusion), and generative design platforms (e.g., NVIDIA GauGAN, Runway ML), have experienced significant advancements, showcasing exceptional capabilities in content generation and data analysis. These technologies have found extensive

applications across creative industries, including art [11], design [17], and literature [18], fundamentally transforming workflows and enabling highly personalized and dynamic creative processes [2]. In the context of education, recent studies have also highlighted the growing role of GenAI in enhancing creative learning across various educational settings. For instance, Sari et al. explored the use of GenAI in STEM-based creative learning, highlighting its potential to enhance problem-based learning and improve cognitive, affective, and psychometric outcomes for educators and students [48]. Research from Lim et al. found that AI, as a relational artifact, enhances creativity by engaging students in collaborative artistic exploration [32]. Suh and An studied how GenAI enhances creative learning, enabling students to create comics by linking code to stories and visuals in a visual programming environment [51]. However, current research in this field either focuses on formal educational settings or addresses only specific aspects of creative learning, such as ideation (e.g., generating prompts or inspiration [21, 25, 39]) and creation (e.g., providing technical support [11, 35]), often overlooking the broader context of the cyclical creative learning journey. Resnick’s creative learning strategies, inspired by kindergarten-style learning and applicable to learners of all ages, highlight an iterative cycle—Imagine, Create, Play, Share, Reflect—that mirrors the recursive nature of creativity [46]. While GenAI technologies have shown promise in delivering personalized guidance [3, 23] and real-time feedback [15, 44] throughout various phases—from ideation [13, 28, 50], creation [1, 33], to reflection [4, 29]—they have not yet been fully integrated to offer dynamic support at each stage in the creative learning cycle. Additionally, the application of GenAI in creative learning of nonformal educational settings, such as museums, remains underexplored. Unlike formal classrooms, where structured curricula and teacher guidance are integral, informal learning environments demand more autonomous, self-directed engagement [6]. These gaps highlight the untapped use of GenAI to support the full, cyclical process of creative learning, especially in nonformal settings. There is a clear need for platforms that combine AI’s generative capabilities with features designed to support the exploratory, self-directed nature of creative learning.

3 STUDY CASE AND DESIGN GOALS

We selected maritime art painting creation as a study case for exploring how GenAI can be integrated into creative learning support systems for museum education for several reasons. First, painting is a widely recognized form of creative learning in museum education. Second, maritime art plays a key role in preserving and communicating the historical, cultural, and technological advancements related to human exploration and maritime trade, making it an essential part of museum collections. Third, maritime art painting is accessible, featuring vivid imagery of ships, the ocean, and maritime life, which makes it relatively easy to understand and requires minimal cultural interpretation. We tested the performance of GenAI in creating maritime art and responding with knowledge in this domain. We found the results to be highly reliable, likely attributable to the extensive training datasets, including a substantial volume of relevant artistic works and text materials. Building on Resnick’s iterative creative cycle of five stages: —Imagine, Create, Play, Share, Reflect [46], we have identified five interlinked design goals (DG).

While each goal naturally aligns with a primary phase in Resnick’s cycle, the DGs also support the overall, looping process.

DG1 Inspire through Contextual and Interactive Exploration (“Imagine”). According to constructionism [8] and Resnick’s emphasis on the “Imagine” phase, learning is most effective when individuals actively construct knowledge through meaningful, open-ended exploration. The system should provide rich historical and contextual materials supported by curated resources on maritime art history and culture to inspire users. Interactive learning, enhanced by LLMs, should be integrated to facilitate idea exchange and offer real-time, adaptive guidance. This also supports later stages by naturally evolving users into sketches (Create) and fueling experimentation (Play).

DG2 Enable Seamless Creative Expression (“Create”). Creative learning thrives on active participation, where transforming ideas into tangible outcomes deepens understanding and encourages exploration. Research emphasizes the importance of interactive, intuitive tools that enable learners to freely express and refine their creativity [16]. The system should offer supportive tools and AI-driven guidance, enabling users of all skill levels to engage fully in creation based on the inspiring content in the Imagine stage. To reduce barriers and enable seamless expression, GenAI should be leveraged to alleviate technical challenges for users and further encourage refinement through creation and feedback.

DG3 Support Iterative Experimentation (“Play”). The “Play” phase in Resnick’s creative learning strategies emphasizes experimentation and exploration, where low-stakes environments empower learners to test ideas and refine through iteration [38]. The system should provide GenAI capabilities for generating images and text to foster an environment where users can freely explore and experiment with various artistic ideas, such as elements and narratives, making the creative process both engaging and rewarding.

DG4 Facilitate Sharing and Peer Learning (“Share”). Sharing is a vital part of the creative process, with peer interaction and knowledge exchange broadening horizons and building a sense of community [24]. The system should provide welcoming and supportive spaces that encourage users to share work created in the prior learning stages and learn from each other. GenAI can enhance this process by generating shareable materials to facilitate interaction and engagement.

DG5 Cultivate and Encourage Reflective Practices (“Reflect”). Reflection, a cornerstone of learning, transforms learning experience into actionable knowledge [12]. In creative learning, encouraging users to evaluate their creative decisions and growth is vital for meaningful outcomes. The system should support reflection by enabling users to document their creative process in previous stages and receive personalized feedback while also prompting reflective conversations with GenAI to deepen understanding and sustain engagement.

4 THE MARISCOPE SYSTEM

The platform Mariscope we have designed aims to foster creative learning in maritime art through an interactive, GenAI-supported environment. It enables a personalized, dynamic journey where

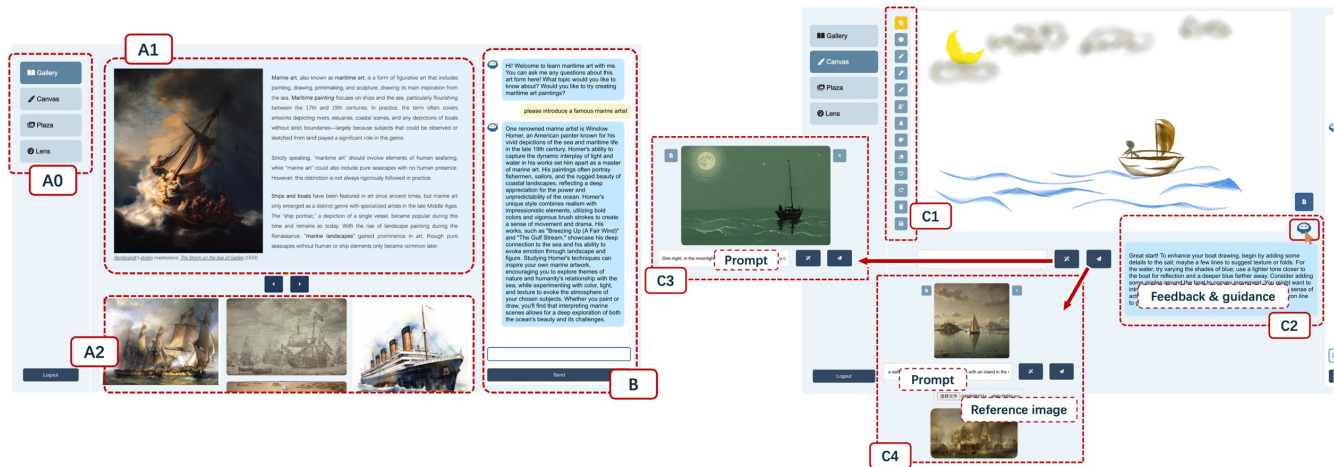


Figure 2: The user interface of the Gallery (left) and Canvas (right) views, including features for exploring maritime art through curated content (A), a persistent interactive chatbot for contextual learning (B), and a dynamic drawing environment with AI-driven enhancements (C).

users learn about maritime art while actively creating and refining their artistic expressions through a cyclic creative process. Mariscope is a web-based application accessible via desktop browsers and the system comprises four views—Gallery, Canvas, Plaza, and Lens—along with a persistent LLM-based Chatbox with a chatbot to guide users throughout their learning journey. Mariscope utilized GPT-4 as the underlying LLM and leveraged the Stability AI API service for all image generation tasks. All prompts used in Mariscope are shown in Appendix A.

4.1 Exploring Through Context, Inspiration, and Interactive Learning Support (DG1)

The Gallery view (Fig. 2-left) serves as the entry point for users to explore the cultural and historical context of maritime art. It features curated content pages (Fig. 2-A1), providing insightful introductions to maritime art, and an Art Gallery Section (Fig. 2-A2), showcasing a selection of prominent maritime artworks. These organized materials aim to deepen users' understanding of maritime art, providing both contextual knowledge and creative inspiration. Alongside the Gallery, the system integrates a persistent Chatbot (Fig. 2-B) designed to provide continuous and interactive engagement throughout all stages. We used the GPT-4 Chat Completion API to develop this chatbot, with a tailored prompt to interact with users. As a friendly learning companion, the chatbot greets users warmly, encouraging exploration of maritime art, its cultural and historical context, and artistic techniques. It also guides reflection on gallery artworks and inspires users to create their own artistic projects.

4.2 Crafting with Interactive Drawing, Real-Time Feedback and AI-Driven Enhancements (DG2)

The Canvas view (Fig. 2-right) provides users with a dynamic, interactive drawing environment where they can bring their ideas to life.

The canvas offers a variety of brushes, essential drawing tools and basic operations to create a maritime art painting (Fig. 2-C1). There is a painting-guiding chatbot in the bottom right corner of the canvas (Fig. 2-C2), providing guidance and creative suggestions with an encouraging prompt to help users initiate their artistic process. Powered by the GPT-4 Vision feature, the chatbot delivers real-time feedback when users click on it, offering actionable suggestions and positive reinforcement to help refine their drawings (Fig. 2-C2). Under the canvas, two optimization features leverage Stability AI's ControlNet Sketch and Style API to enhance the user's sketch or generate a customized reference image. Users can describe their current drawing in text and input it as a prompt, allowing the AI to generate an enhanced artwork that aligns with the prompt while maintaining the composition of the sketch on the current canvas (Fig. 2-C3). To generate a customized reference image, users can upload an existing image and provide a textual prompt for the tool to generate professional artwork based on the style of the reference image and the theme in the prompt (Fig. 2-C4). Both user-created artworks from the canvas (hand-drawn by the user) and AI-generated pieces based on the user's painting can be uploaded to the Plaza as part of the user's portfolio.

4.3 Experimenting, Sharing and Connecting Leveraging AIGC and Creative Iteration (DG3&4)

In the Plaza view (Fig. 3-left), users can upload their creations to either a private "My Works" section or a public "Square" for others to view (Fig. 3-D1). Users are encouraged to explore and try out various iterations of their or others' artworks by experimenting with different elements and stories with a prompt (Fig. 3-D2). For instance, users can prompt the tool to reimagine their artwork by altering key elements (e.g., replacing a ship with a shark in a seascape). The updated artwork, generated in seconds via Stability AI's ControlNet Style API, is accompanied by GPT-4-generated



Figure 3: The user interface of the Plaza (left) and Lens (right) views, including features for experimenting with artwork through prompts, sharing creations via GenAI-assisted storytelling, and reflecting on progress with constructive feedback and guided prompts.

contextual insights on the significance of the elements' significance in maritime art traditions. Before publishing their creation to the "Square," users are encouraged to write a brief story (Fig. 3-D3) to share their creative process, including their intentions and inspiration. AI assistance powered by GPT-4 Chat Completion API is available to polish the text and add details, with users retaining the ability to edit and refine the AI-generated content before finalizing their publication. All the published pieces are displayed in the "Square" section (Fig. 3-D4); users can engage with other users' artworks, viewing their paintings and stories (Fig. 3-D5) and exploring others' creations in the same manner as for their creations.

4.4 Reflecting through AI-driven Feedback, and Guided Prompts (DG5)

The Lens view allows users to reflect on their own creative journey. As users create their maritime artworks, they save any versions of their creations by simply clicking a "record" button next to the Canvas (Fig. 2-C1). These images are then stored in the Lens (Fig. 3-E1) for users to reflect on and compare different stages of their creative process. Users can click on the button under each piece to receive constructive feedback generated by GPT-4 Vision (Fig. 3-E2), which offers tailored comments on techniques and suggestions, helps users assess their choices, deepen their understanding of maritime art, and foster growth. Additionally, the Lens features a series of reflection prompts related to the creation of maritime art (Fig. 3-E3). These prompts are designed to provoke deeper thought about the user's artistic decisions and learning process. Users can refresh to receive more reflection questions, and when they select a question, the chatbot responds to guide them through the reflection process (Fig. 3-E4) and offers suggestions for improvement.

5 PRELIMINARY USER STUDY

We conducted a preliminary evaluation to investigate the usability and effectiveness of the platform. The study aimed to measure participants' learning outcomes, creative confidence, and support

received during creative learning, particularly the user experience of integrating AI in creative learning.

5.1 Participants and Procedure

We recruited four participants (P1–P4, 2 male and 2 female) without prior experience in art or maritime art, ensuring the evaluation targeted non-expert engagement and education. Participants, aged 25 to 35, came from diverse professional backgrounds. The study began with a pre-questionnaire assessing their background knowledge of maritime art and painting experience using a 5-point Likert scale. This was followed by a 30-minute face-to-face or online onboarding session to introduce the platform's goals, structure, and features and 10 minutes for independent exploration. Over two days, participants accessed Mariscope via a link and used their computers (laptops or desktops) in their natural environments, publishing at least one artwork in the "Square." A post-questionnaire then asked questions on user experience and the same questions in the pre-questionnaire to assess knowledge acquisition and creative self-efficacy, also on a 5-point Likert scale. Finally, semi-structured interviews provided qualitative insights into their learning process, AI interactions, and suggestions for improvement. The questionnaires were adapted from surveys used in studies of creative learning and art creation in museums [22, 52].

5.2 Findings

Over the course of two days, participants explored the platform at their own pace. They engaged with the Gallery and Chatbox to learn about maritime art, produced a total of six hand-drawn sketches on the Canvas, created eleven polished AI-generated pieces using GenAI functions, and ultimately posted seven artworks (only one hand-drawn sketch) in the Plaza's sharing space. Although the sample size is small (N=4), this exploratory data offers initial insights into how non-expert learners engage with GenAI-facilitated creative experiences in digital museum contexts.

5.2.1 Knowledge acquisition and creative self-efficacy. The findings from the pre- and post-questionnaire revealed remarkable improvements in both knowledge acquisition and creative self-efficacy. Compared to before the experiment, all participants reported a deeper understanding of maritime art, with knowledge scores rising by an average of two points (on a five-point scale). Three participants, in particular, highlighted an enhanced familiarity with common elements and symbols in maritime art. All participants felt more confident in their ability to create maritime artwork than before. In terms of creative self-confidence, three participants expressed greater confidence in their artistic abilities after engaging with the platform and they also reported being able to create better artwork and propose more innovative ideas than before, reflecting an increase in their confidence in creative tasks. However, one participant (P3) mentioned that he felt overly reliant on GenAI features, which made him uncertain about the improvement of his creative abilities.

5.2.2 Supports received during creative learning. All participants valued the curated contents and professional artworks in Gallery view for learning and inspiration. P1 and P4 found detailed introductions and refreshing content motivating, especially during creative blocks. Three participants noted the Chatbox provided interactive learning and targeted guidance, enhancing inspiration and understanding. P2 and P4 praised its clear and immediate support during the learning and creating. Every participant agreed AI's ability to refine drafts into polished works significantly boosted their creative expression, though P3 and P4 desired more precise adjustments, feeling the AI sometimes misunderstood their intent. Real-time guidance was universally appreciated, with P1 and P4 suggesting single-step instructions for better accessibility, particularly for those without an artistic background. All participants explored the experiment function in Plaza's play stage, with P2 and P4 finding it engaging as adding elements generated new visuals and reinforced learning. However, P3 felt the play stage lacked impact, describing it as merely a generation feature with little interactivity or playfulness. In the sharing stage, participants all enjoyed browsing others' work and were encouraged to share their own creations. Participants (P1,2,3) appreciated the AI-generated story but preferred shorter, more concise descriptions, leaving room for personal interpretation of artworks. In the Lens (for reflection) section, participants have reflected on their historical creations through this feature, but the depth and engagement of their reflections varied. P2 found pre-formulated reflective questions useful, as they offered a structured way to think about improvements. P3 suggested that feedback could include data visualizations and statistical insights for a more objective and engaging evaluation. Additionally, P1 preferred feedback that was segmented into clear points for easier comprehension.

5.2.3 System usability and user experience. All participants reported the system is easy to learn, and most (three) of them rated the platform as easy to use and intuitive, feeling confident in their ability to use the platform effectively with minimal technical difficulties. They highlighted that the responsiveness of the system made the experience enjoyable and easy to navigate. However, the other participant (P1) reported difficulty navigating some AI functions, suggesting clearer titles and subtle animations to enhance

clarity and visual appeal. This underscores the need for simpler instructions, more guided interactions, and streamlined feature labeling in future enhancements of the system to better accommodate diverse user preferences. All participants appreciated AI's assistance in guiding their creative process and three users stated that collaborating with GenAI during the creative learning process enhanced the creativity and completeness of their works. Notably, all users expressed a desire for more museums to adopt similar online learning platforms.

6 DISCUSSION

In this section, we synthesize the study's main findings and consider how GenAI-supported creative learning enhances digital museum experiences. We also note challenges, future directions, and broader implications for nonformal educational settings.

6.1 Systematic and Interactive Learning in Digital Museum Education

In Mariscope, we have digitally and intelligently integrated the complete creative learning path to offer a systematic and interactive learning experience for digital museum education. Participants appreciated the combination of museum learning and creating, which distinguished the platform from traditional digital museums by enabling digital hands-on exploration alongside education. They also reported feeling supported and free from confusion throughout this interactive learning process, from gaining knowledge to creating and sharing their works. Yet, we found that participants required longer learning cycles and deeper engagement in the reflection stage, supported by objective insights such as data-driven learning analytics. Additionally, they expressed a need for more diverse external resources during the Imagine and Play phases to enrich their creative exploration and experimentation. For future steps, we will enhance the platform by integrating robust data-driven insights to support reflective practices and incorporating a broader range of learning materials and tools to inspire and empower users across all stages of the creative process.

6.2 Human-AI Co-creation for Non-expert Learners in Informal Educational Settings

Feedback from participants was generally positive regarding the integration of AI's generative and adaptive capacities into the creative learning path. They all reported that, to varying degrees, AI's feedback and enhancements during the creation process made their journey more inspiring and engaging, allowing them to experience the joy of creation even without professional artistic knowledge or skills. However, they noted that AI could sometimes generate uncontrollable content and expressed a desire for more precise control over AI-generated results during co-creation. Additionally, they emphasized the need for concise and clear AI guidance that is easily accessible to non-expert learners without requiring professional-level understanding. In the future, we plan to integrate parametric adjustments into the Gen-AI functions to enable more precise refinement over AI-generated content according to individual intents, empowering users in co-creation while improving the clarity and accessibility of AI guidance for non-experts.

7 LIMITATION AND FUTURE WORK

We acknowledge that the small sample size significantly limits the generalizability of our findings and confines our ability to draw strong conclusions. Future studies will involve larger participant groups to further validate the platform's effectiveness and insights. Currently, Mariscope is a web-based application to ensure easy access and broad compatibility. In future iterations, we plan to adapt the platform for tablets to facilitate direct stylus input and enhance the hands-on nature of the drawing tools. Additionally, all introductions to maritime art were generated using a large language model (LLM). While testing and experiments showed no issues, with all outputs working effectively and making sense, the potential risk of AI hallucination still exists. Future work will focus on enhancing mechanisms to ensure the quality and accuracy of generated content, both text and images. This includes pre-training the LLM with domain-specific knowledge and datasets related to maritime art, providing a solid foundation of accurate and contextually relevant information.

8 CONCLUSION

In this work, we presented Mariscope, a GenAI-enabled platform that integrates the entire creative learning cycle—Imagine, Create, Play, Share, and Reflect—into digital museum education. A preliminary evaluation with four participants suggests positive impacts on learning outcomes and creative self-efficacy, though further user testing is needed. Future work will improve AI controls, expand diverse resources, and transition to tablet-based interactions, aiming to offer a more responsive and data-informed experience for creative learning in digital museums.

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A PROMPTS FOR CALLING LLM

A.1 Chatbot Conversation

You are a mentor specializing in the cultural history and artistic creation of maritime art. Please provide professional answers to the following questions, along with relevant knowledge and creative guidance.

A.2 Real-time Feedback and Guidance for Users' Drawings

You are a maritime art painting mentor. This image represents the work of a student who is just beginning to learn maritime art painting and now needs your guidance to further develop the piece. Based on the content of the image, provide step-by-step instructions in second person to guide the student in advancing their creation.

A.3 Introduction for Experimental Artistic Elements

You are a maritime art mentor. Based on the following content: (User's prompt), please introduce relevant knowledge about maritime art painting related to the mentioned artistic elements. Provide insights into this type of element in maritime art, such as its cultural significance, context, and historical stories.

A.4 Text Refinement and Story Enhancement

This is a story introducing a maritime art painting. Please refine the story to make the language more fluent and enhance its vividness and appeal.

A.5 Feedback on Previous Work

You are a maritime art painting mentor reviewing the work of a student who is just beginning to learn maritime art painting. Provide a professional critique in a warm and encouraging tone, focusing on constructive suggestions and guiding the student to reflect on their work.

A.6 Reflective Conversation

You are a mentor specializing in maritime art painting. A student learning maritime art painting seeks to reflect on their work with the following question: (user-selected question). Please guide them in exploring creative reflections and thoughts on this question, interacting with them through thoughtful prompts and questions.